# SALES INSIGHTS VISUALIZATION PROJECT

Tools: SQL + Power BI

# BUSINESS OBJECTIVE

- A computer hardware manufacturer, Atliq Hardware, is struggling to track sales and gain clear insights in a dynamically growing Indian market
- The Sales Director is frustrated by verbal reports from regional managers that often "sugarcoat" the truth and Excel files that are too massive and complex to provide digestible insights
- The project focuses on building a Power BI dashboard that converts raw data into visual trends, allowing the director to see the "truth" behind the numbers without relying on biased verbal feedback.

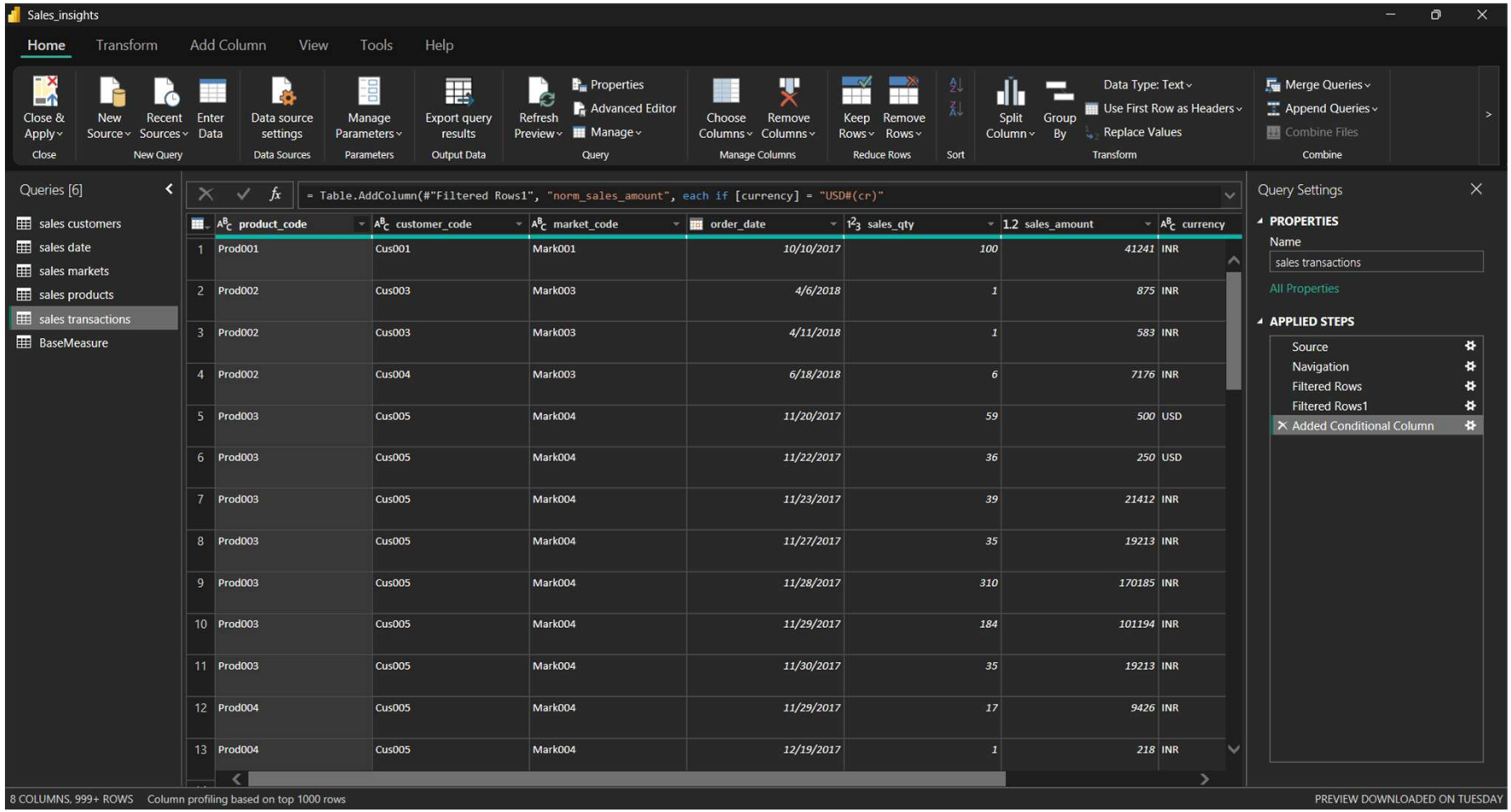
# PROBLEM STATEMENT / PROJECT SCOPE

- Identify and analyze sales data sources related to business performance
- Overcome challenges of unreliable verbal reports and complex Excel data
- Clean, transform, and model the data for effective analysis
- Develop an interactive sales dashboard to track key KPIs
- Provide filters to slice and analyze data across different dimensions
- The dashboard should deliver both high-level overview and detailed insight

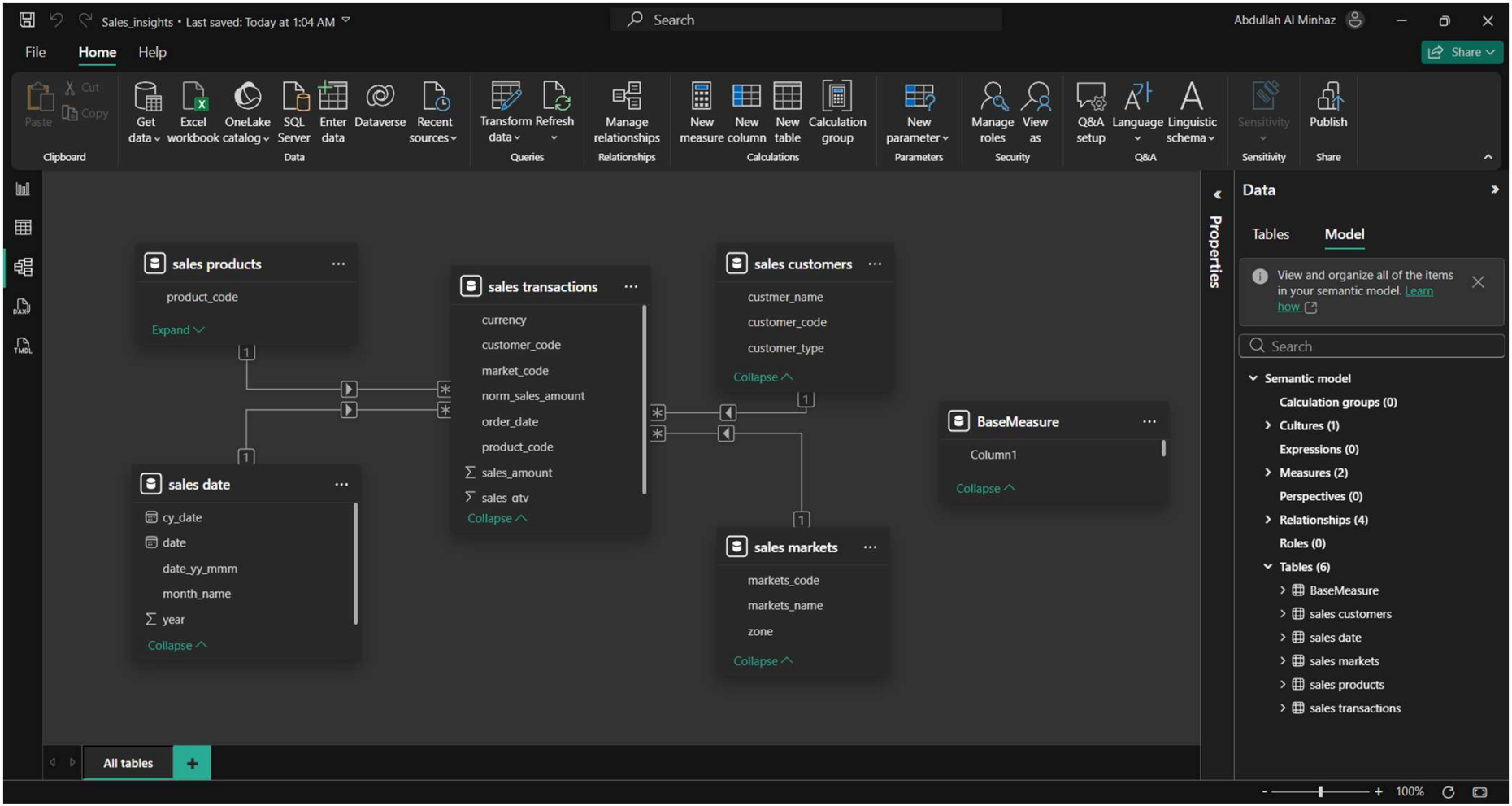
# SOLUTION APPROACH

- There are 5 tables provided for tracking revenue, 3dimension tables (Customers, Products, Markets, Date) and 1 fact table (Transactions)
- Power BI was the tool used for creating the visualization/dashboard
- The data was imported, analysed and transformed as per necessity within Power Query
- The relationships between the tables were created within Power Pivot

# DATA CLEANING/TRANSFORMATION IN POWER QUERY



# DATA MODELLING IN POWER PIVOT

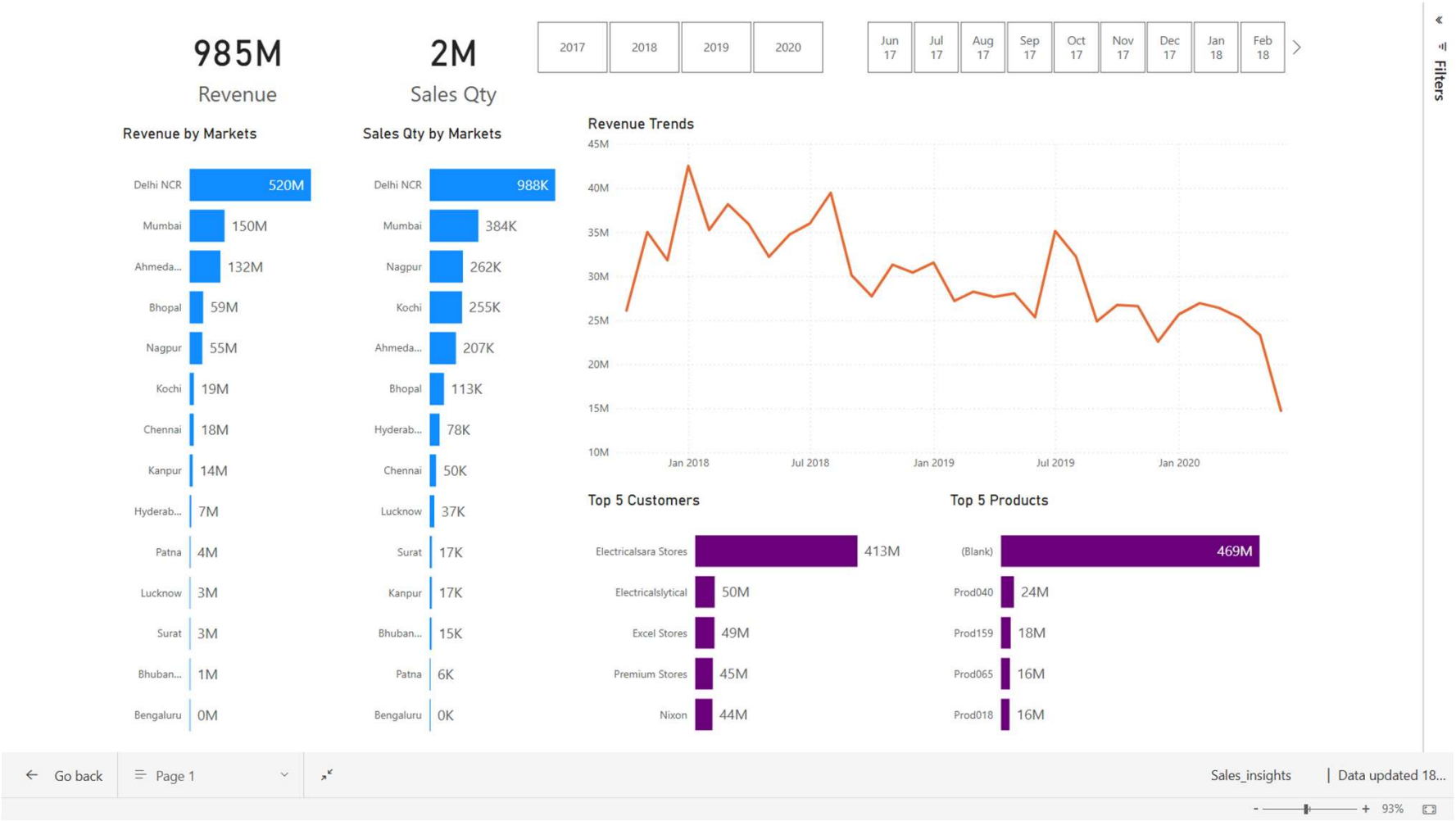


# SOLUTION APPROACH

A few measures were created to calculate the KPIs as shown below:

- **Revenue** = SUM(transactions[sales_amount])
- **Sales Quantity** = SUM(transactions[sales_qty])
- **Revenue Growth (YoY)** =
- (Current Year Revenue – Previous Year Revenue) / Previous Year Revenue
- **Top Customers** = Based on highest revenue contribution
- **Top Products** = Based on highest sales/revenue

# REVENUE DASHBOARD



# FEATURES OF THE DASHBOARD

The following 5 visuals were provided:

- **Revenue Overview Cards**

  Displays total revenue (985M) and total sales quantity (2M)

- **Revenue by Markets**

  Bar chart showing revenue contribution across different regions

- **Sales Quantity by Markets**

  Visual comparison of sales volume across markets

- **Revenue Trend Analysis**

  Time-series line chart showing revenue trends from 2017–2020

- **Top 5 Customers**

  Highlights key customers contributing highest revenue

# FEATURES OF THE DASHBOARD

- The following filters were provided to slice and dice the data:
  - Month-year
  - City
  - Top Customers
  - Top Products
- The visuals are interactive in nature RP1
- Tooltips pop-up when hovering over a visual for more information about the data point

## Slide 10

**RP1** Raghavan P, 12/26/2022

# BUSINESS OUTCOMES

The following key business insights were derived from the sales dashboard:

- Delhi NCR contributes the highest revenue among all markets, while some regions show significantly lower performance, indicating areas for improvement
- A small group of customers contributes a major portion of total revenue, highlighting the importance of key client relationships
- Certain products generate higher revenue compared to others, helping identify best-performing product categories
- Revenue trends show fluctuations over time, indicating seasonal patterns and potential market dynamics

# BUSINESS OUTCOMES

The following key business insights were derived from the sales dashboard:

- Some markets have low sales quantity and revenue, suggesting opportunities for targeted business strategies
- The dashboard enables identification of high-performing and underperforming regions for better resource allocation
- Data-driven insights support strategic decision-making for business growth
- The implementation of this dashboard can potentially lead to an estimated revenue increase of around 7%

# CONCLUSION

- Developed a centralized Power BI dashboard to replace complex Excel reports and unreliable verbal updates
- Used SQL for data exploration and Power BI for ETL, data modeling, and visualization
- Provided clear, interactive insights for tracking sales performance
- Enabled data-driven decision-making with a single source of truth

# THANK YOU!